Chapter 5

Consumers and Incentives

ECON 003
UC Riverside · Summer 2026



Where We Left Off

Chapter 4 handed you the demand curve as a fact.
Chapter 5 builds it out of one person making one decision.

CHAPTER 3 GAVE YOU THE TOOL



Optimisation. Compare marginal
benefit with marginal cost and stop where they meet.

CHAPTER 4 GAVE YOU THE PICTURE



Two curves crossing. Demand sloped down.

CHAPTER 5 JOINS THEM UP



A buyer with tastes, prices and a budget. Her choices ARE the demand curve.

Today's Roadmap

01 **The Buyer's Problem**

Three ingredients: what you like, what it costs, what you have

02 **Putting It All Together**

Marginal benefit per dollar — the rule that picks the best bundle

03 **From the Problem to the Demand Curve**

Change one price, redo the problem, and the curve appears

04 **Consumer Surplus**

What buyers get out of a market over and above what they pay

05 **Demand Elasticities**

Putting a number on 'how responsive' — and why it decides revenue

A Question to Carry Through the Hour

Would a smoker give up the habit for $100 a month?

THE INSTINCT

"No. Addiction isn't a budgeting problem — money won't fix it."

"Or maybe yes, if they're broke enough."

Both are guesses. Neither puts a number on anything.

HOLD ONTO IT

We come back to this in Part 4, once you can put $100 into a budget constraint and read off what it buys.

There is a real experiment behind it, and the answer is a number.



The habit, priced

Image: Acemoglu, Laibson & List, Microeconomics 3e

PART 1

The Buyer's Problem

The Buyer’s Problem

Every purchase you have made answered the same three questions. Economists call the three together **the buyer's problem.**

WHAT DO YOU LIKE?



Tastes and preferences. Taken as given and read off your choices.

HOW MUCH DOES IT COST?



Prices. Set by the market in Chapter 4.

HOW MUCH MONEY DO YOU HAVE?



Your budget constraint.

Drop any one of the three and the theory gives nonsense — a private jet, or the cheapest thing on the shelf.

What You Like: Your Choices Give You Away

We do not need to ask what you value. We watch what you buy.

THE $50 JEANS

You spend birthday money on a $50 pair of jeans at the mall.
We now know more than "you like jeans."

WHAT WE ACTUALLY LEARN

Of everything in that mall you could have had for $50 — shoes, dinner, headphones, you favored a pair of jeans over all of them.

So: what do your buying decisions signal about you as a consumer?

Two working assumptions: you want the biggest bang for your buck, and what you buy shows what you like.

Assumptions

PRICES

1. Prices are posted, not negotiated.

2. You can buy as much as you want without pushing the price up.

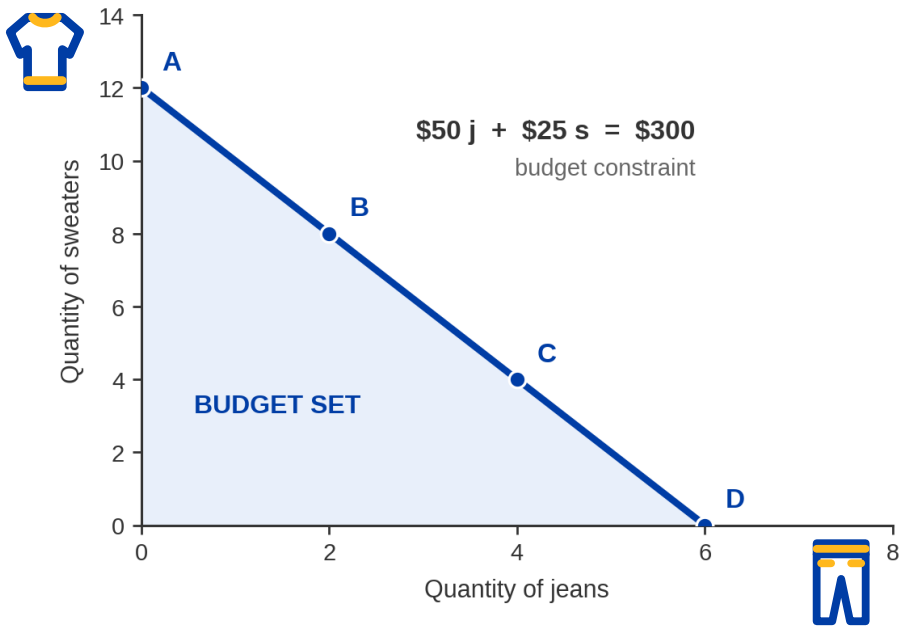
You are one buyer among millions — too small to move the market you met last week.

BUDGET

1. No saving, no borrowing. Money not spent is simply lost.

2. We draw a straight line, but you buy whole units — two sweaters, not 2.4.

Exhibit 5.1 — The Budget Set and the Budget Constraint



FOUR BUNDLES ON THE LINE

Bundle	Sweaters	Jeans
A	12	0
B	8	2
C	4	4
D	0	6

$300 to spend.

Sweaters $25 · Jeans $50.

This is not a demand curve. It plots one good against another — and the prices are hidden in the slope.

One Line, Three Regions



ON THE LINE

Spends exactly $300. These are the bundles worth arguing about.

INSIDE

Affordable, but leaves money unspent — and unspent money is wasted here.

OUTSIDE

Unaffordable. No borrowing, so these bundles simply do not exist for you.

An optimizing buyer always ends up ON the line. Inside is leaving money on the table; outside is impossible.

The Slope Is an Opportunity Cost

slope of the budget constraint = − P(jeans) / P(sweaters) = − $50 / $25 = − 2

MOVING RIGHT ALONG THE LINE



One more pair of jeans costs you TWO sweaters. That is the opportunity cost of a pair of jeans.

MOVING LEFT ALONG THE LINE



One more sweater costs you HALF a pair of jeans. Same trade, read the other way round.

Opportunity cost is measured in units of goods of A per unit of good B you give up (e.g. sweater per pair of jeans), or a price ratio. Jeans cost twice what sweaters cost, so jeans cost two sweaters.

Chapter 3 told you every choice has an opportunity cost.

PART 2

Putting It All Together

Somebody Offers You a $100,000 Car

MONDAY

A billionaire offers to buy you a Jaguar — a $100,000 car.

Do you accept?



TUESDAY

He hasn't got time to go to the dealership, so he offers you a $100,000 check instead.

Do you go and buy the car?



Almost everyone says yes on Monday and no on Tuesday.
Same $100,000 either way — so the difference is not the money.

Given the cash you buy something else, because something else delivers more benefit per dollar. That is the rule.

Exhibit 5.2 — Your Buyer's Problem, with $300

Qty	Sweaters $25 Total benefit	Marginal benefit	MB per $ (÷ $25)	Jeans $50 Total benefit	Marginal benefit	MB per $ (÷ $50)
0	0	—	—	0	—	—
1	100	100	4.00	160	160	3.20
2	185	85	3.40	310	150	3.00
3	260	75	3.00	410	100	2.00
4	325	65	2.60	490	80	1.60
5	385	60	2.40	520	30	0.60
6	435	50	2.00	530	10	0.20
7	480	45	1.80	533	3	0.06
8	520	40	1.60	535	2	0.04

To find the optimum: STEP 1 – Difference to get marginal benefits.

STEP 2 – Divide by goods price to get MB per $ for each goods.

The optimum is 6 sweaters and 3 jeans: 6 × $25 + 3 × $50 = $300, for $435 + $410 = $845 of benefit.

Diminishing Marginal Benefit

Diminishing marginal benefit — each additional unit of the same good is worth less to you than the one before it.

THE FIRST ONE

The first sweater is worth 100 to you. It fills the biggest gap you had.



THE FIFTH ONE

Worth 60. Still positive — you would take it free — but it is not the sweater you needed.



PUSH FAR ENOUGH

Marginal benefit goes to zero, and then negative. An eighth sweater becomes a nuisance.



Marginal Benefit Per Dollar

marginal benefit per dollar = marginal benefit ÷ price

COMPARING RAW BENEFIT IS WRONG



The 1st pair of jeans is worth 160;
the 1st sweater only 100.

So buy jeans first? No — the jeans
cost twice as much.

COMPARING PER DOLLAR IS RIGHT



Jeans: 160 ÷ 50 = 3.20
Sweater: 100 ÷ 25 = 4.00

The sweater delivers more benefit
for every dollar. Buy it first.

Unify the units of benefits of different goods to the per dollar benefit.

Working It: Buy in Order of Bang per Buck

Take the highest number in either MB-per-dollar column, buy that unit, cross it off, repeat until the $300 is gone.

Step	Buy	MB per $	Cost	Spent so far
1	Sweater #1	4.00	$25	$25
2	Sweater #2	3.40	$25	$50
3	Jeans #1	3.20	$50	$100
4	Sweater #3	3.00	$25	$125
5	Jeans #2	3.00	$50	$175
6	Sweater #4	2.60	$25	$200
7	Sweater #5	2.40	$25	$225
8	Sweater #6	2.00	$25	$250
9	Jeans #3	2.00	$50	$300

STEP 3: figure out which good should you spend the next dollar on – find the highest MB per $.

STEP 4: Repeat until you exhaust your budget.

The money runs out exactly as the two columns level off at 2.00 — 6 sweaters, 3 jeans.

The Consumer Equilibrium Condition

MB (sweaters) / P (sweaters) = MB (jeans) / P (jeans)

at the optimum: 50 / 25 = 100 / 50 = 2.00

$$\frac{MB_s}{P_s} = \frac{MB_j}{P_j}$$

IF THEY ARE NOT EQUAL

Say sweaters give 3 per dollar and jeans give 2.

You are leaving benefit on the table. Shift a dollar to sweaters and you are better off.

WHY IT SETTLES

Buy another sweater and its marginal benefit falls — diminishing marginal benefit again.

The ratios move toward each other until there is no gain left in switching.

Equilibrium: nobody has any reason left to change what they are doing.

Here: the last dollar you spent give the same benefit regardless of what goods you choose to buy.

A Price Change Pivots the Line

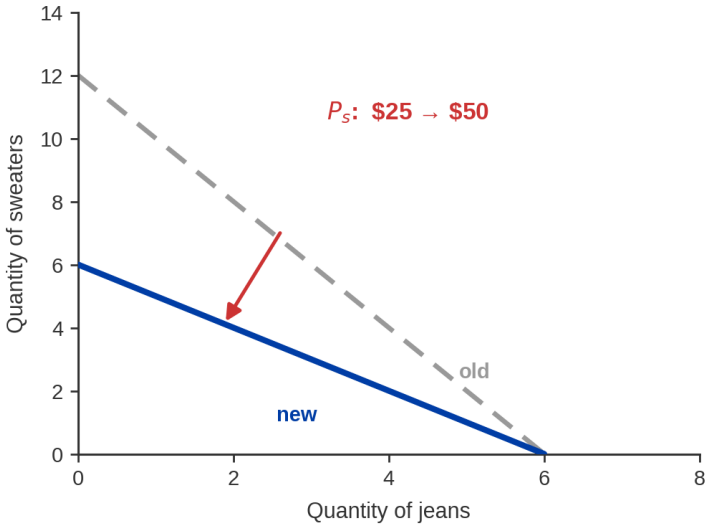


Exhibit 5.3 — sweaters $25 → $50.
Only 6 sweaters now. Pivots on the jeans axis.

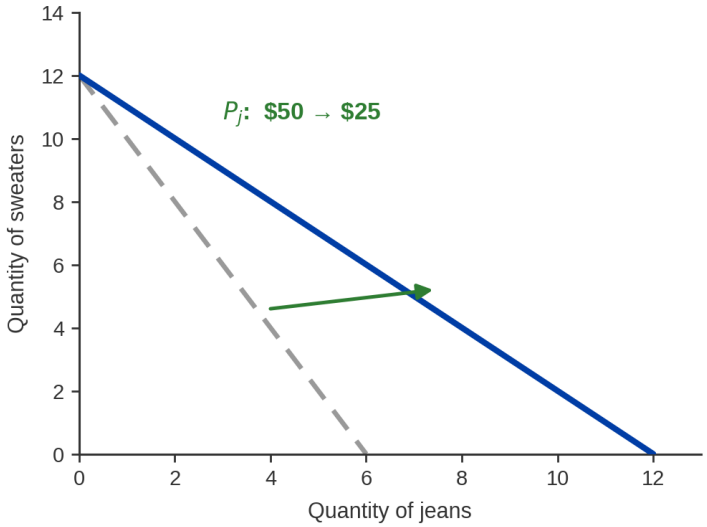
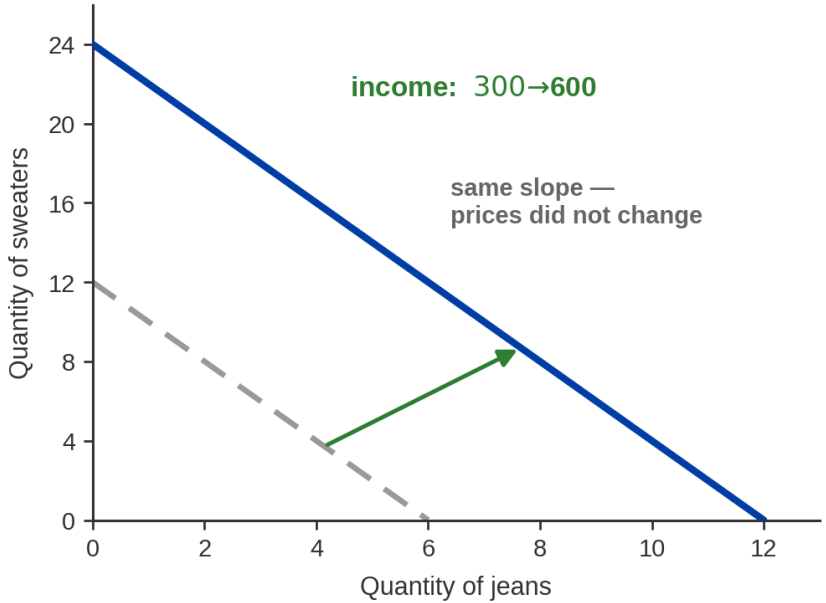


Exhibit 5.4 — jeans $50 → $25.
Now 12 pairs. Pivots on the sweater axis.

Change one price and the line swings: one intercept moves, the other is nailed down. The slope — the trade-off — changes.

An Income Change Shifts It — Without Tilting It



WHAT CHANGED

$300 → $600. Now 24 sweaters or 12 pairs of jeans, or anything between.

WHAT DID NOT

The slope. Prices are untouched, so a pair of jeans still costs two sweaters. Same trade-off, bigger set.

More income buys more of everything. It does not change what one thing costs in terms of another.

PART 3

From the Buyer's Problem to the Demand Curve

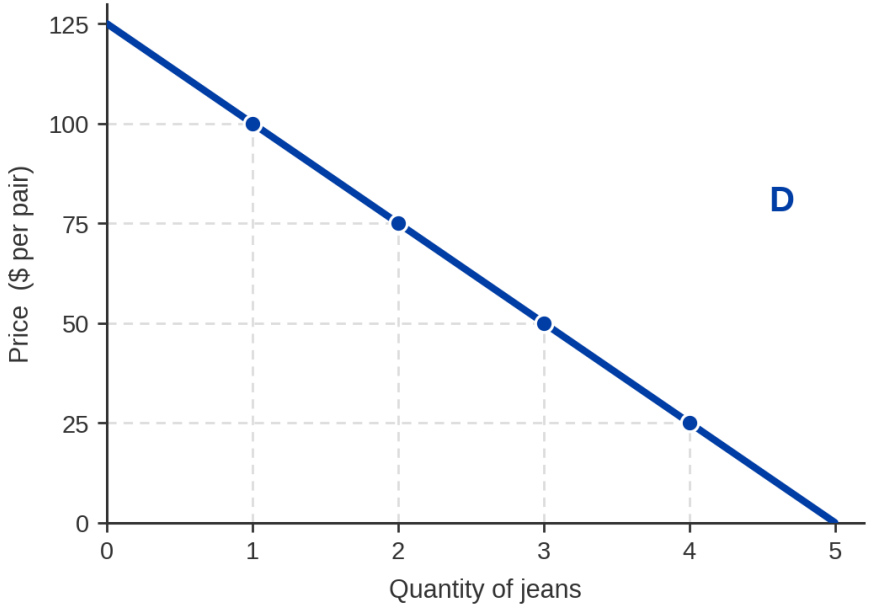
Change the Price, Redo the Whole Problem

Raise the price of jeans and every jeans-per-dollar number falls.
Redo the optimization with budget constraint as price changes.

Price of jeans	MB per $ on the 1st pair	MB per $ on the 2nd	MB per $ on the 3rd	Pairs bought
$25	160 ÷ 25 = 6.40	150 ÷ 25 = 6.00	100 ÷ 25 = 4.00	4
$50	160 ÷ 50 = 3.20	150 ÷ 50 = 3.00	100 ÷ 50 = 2.00	3
$75	160 ÷ 75 = 2.13	150 ÷ 75 = 2.00	100 ÷ 75 = 1.33	2
$100	160 ÷ 100 = 1.60	150 ÷ 100 = 1.50	100 ÷ 100 = 1.00	1

The sweater column never moved. Only jeans got dearer, and only jeans purchases fell.

Exhibit 5.6 — Your Demand Curve for Jeans



FOUR SOLVED PROBLEMS

Price	Pairs bought
$25	4
$50	3
$75	2
$100	1

Plot the pairs. Join them.
That is a demand curve.

Every point on this curve is one full solution of the buyer's problem – under our standard set of assumptions and the alternative's price fixed.

We can use the price of a whole bundle of goods as the alternative to make it more general.



So — Why Does the Demand Curve Slope Down?

Because marginal benefit falls, a lower price is needed to make the next unit worth buying.

START FROM QUANTITY

The 4th pair of jeans is worth less to you than the 3rd. That is diminishing marginal benefit.

SO ASK WHAT IT TAKES

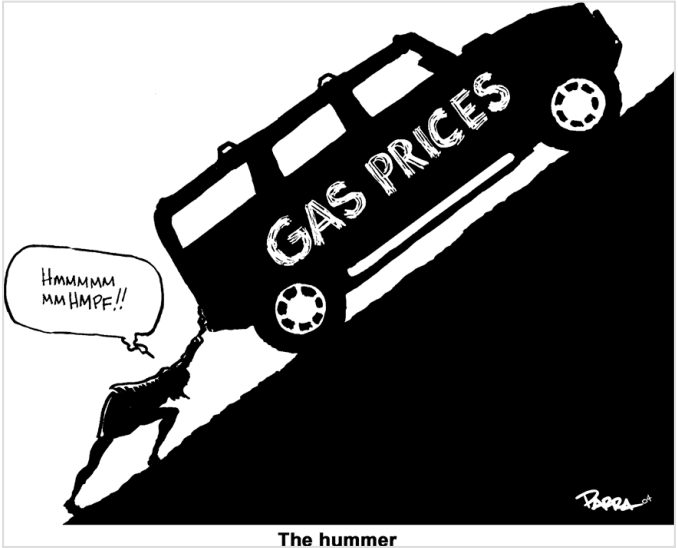
For that 4th pair to clear the Bar, its price per dollar of benefit must improve — the price must fall.

READ IT THE OTHER WAY

Which is the same as saying: when the price falls, quantity demanded rises. The law of demand.

Would Rising Gas Prices Change What You Drive?

A car is a bundle with a running cost attached. Change the cost of running it and you change the bundle.



The hummer

Uphill, on a full tank

WHAT THE FUEL BILL DID

2008 — US regular peaked at $4.11 a gallon in the week to 7 July 2008.

US Hummer sales: 54,052 in 2006 → 20,681 in 2008 → 5,487 in 2009. GM opened a strategic review that June.

Nobody's taste in cars changed in 2008. The price of running one did, and the ranking changed with it.

Image: Acemoglu, Laibson & List, Microeconomics 3e

PART 4

Consumer Surplus

Consumer Surplus

Consumer surplus — the difference between what a buyer is willing to pay for a good and what the buyer actually pays.

WILLINGNESS TO PAY

The most you would hand over rather than go without.

WHAT YOU ACTUALLY PAY

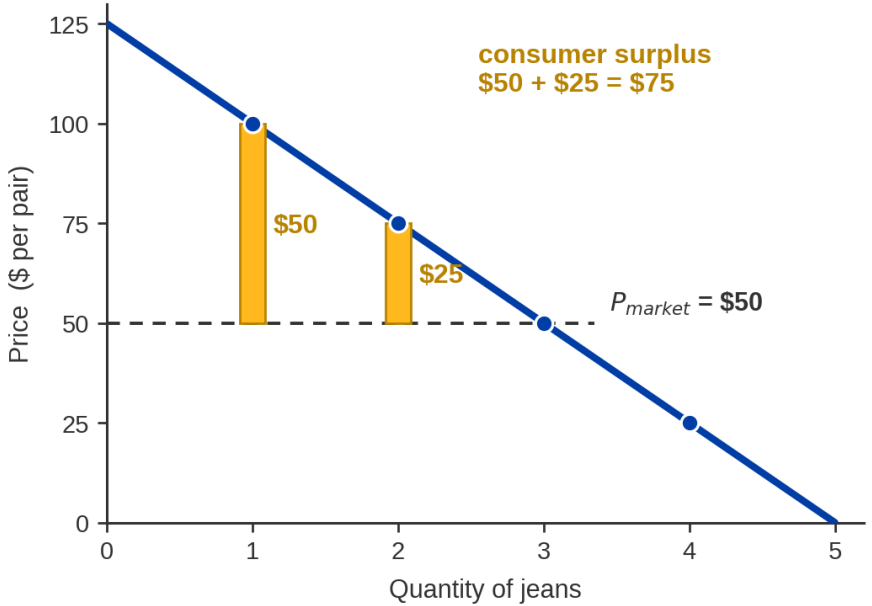
The market price — the same price for everyone, however much they wanted it.

THE GAP IS YOURS TO KEEP

Trade only happens when the gap is positive. Every purchase you make leaves you ahead.

You should only pay for goods that give you consumer surplus.

Exhibit 5.7 — Computing It for One Buyer



AT A MARKET PRICE OF $50

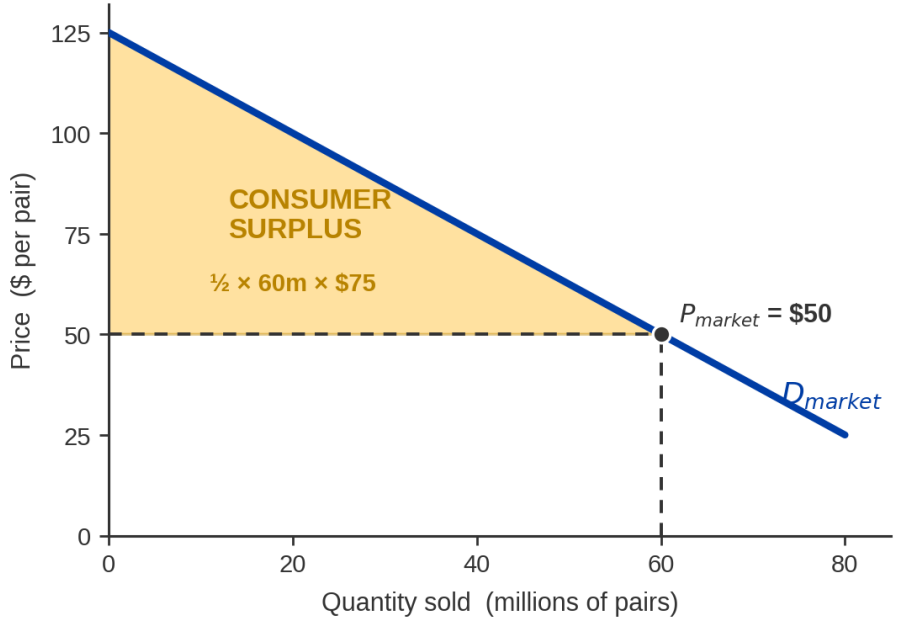
Pair	Worth to you	You pay	Surplus
1st	$100	$50	$50
2nd	$75	$50	$25
3rd	$50	$50	$0

Total consumer surplus

$50 + $25 + $0 = $75

The third pair is the marginal one: worth exactly what it costs, and adding nothing. That is why she stops there.

Exhibit 5.8 — Market-Wide Consumer Surplus



Add every buyer's demand sideways and the steps smooth into one market curve.

CS = ½ × base × height

= ½ × 60 million × $75

= $2.25 billion

It measures what the whole market's buyers gained by being able to trade.

Exhibit 5.9 — What a Price Rise Takes Away



PRICE $50 → $75

Quantity 60m → 40m

CS $2.25 bn → $1.00 bn

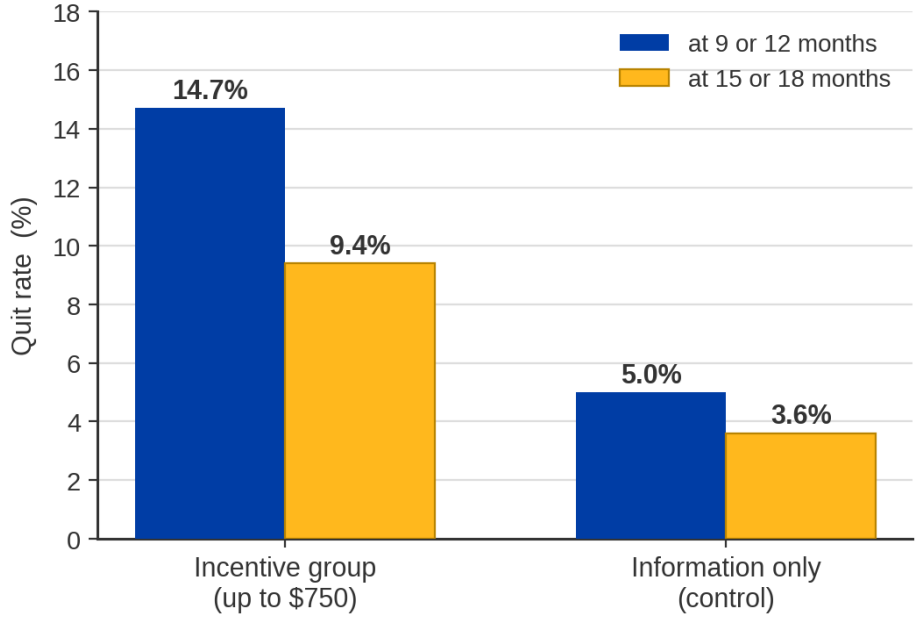
Lost $1.25 billion

WHERE IT WENT

Part goes to sellers as higher revenue. Part is simply gone — buyers priced out of trades they valued.

Price up, surplus down; price down, surplus up. This is how we will judge taxes, ceilings and monopolies later.

Evidence: Would a Smoker Quit for Cash?



THE EXPERIMENT

878 General Electric employees, randomly assigned. One group offered up to $750 for quitting and staying quit; the other got information only.

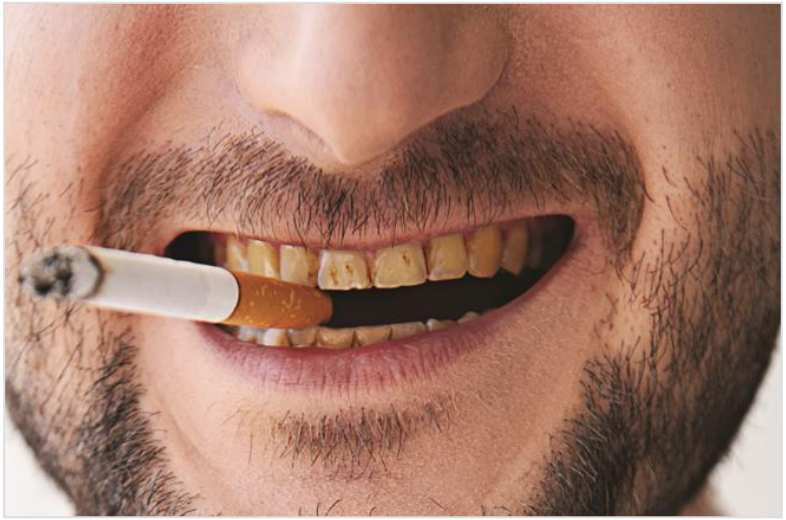
THE CATCH

Roughly three times the quit rate — but nine in ten still smoking, and the gap narrows once the money stops.

Incentives move behaviour in the direction theory predicts. They do not move it as far as anyone hoped.

So Which One Would Actually Make You Quit?

Take a second and answer honestly before we look at what the money does.



What quitting costs you



What quitting pays you

Incentives only work on people for whom the second picture outweighs the first — and the trial says that is a minority.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Your Buyer's Problem, with $400

Quantity	Sweaters $25 Total Benefits (A)	Sweaters $25 Marginal Benefits (B)	Sweaters $25 Marginal Benefits per Dollar Spent = (B)/$25	Jeans $50 Total Benefits (C)	Jeans $50 Marginal Benefits (D)	Jeans $50 Marginal Benefits per Dollar Spent = (D)/$50
0	0	-	-	0	-	-
1	100	100	4	160	160	3.2
2	185	85	3.4	310	150	3
3	260	75	3	410	100	2
4	325	65	2.6	**490**	**80**	**1.6**
5	385	60	2.4	520	30	0.6
6	425	50	2	530	10	0.2
7	480	45	1.8	533	3	0.06
8	**520**	**40**	**1.6**	535	2	0.04

The Answer, in Sweaters and Jeans

An extra $100 a month is not a taste change and not a price change.
It is income — so the budget line shifts out, and the bundle gets bigger.

WITH $300



6 sweaters and 3 pairs of jeans.
Total benefit $845.

WITH $400



8 sweaters and 4 pairs of jeans.
Total benefit $1,010.

SO THE REAL QUESTION IS



Is smoking worth two sweaters and a pair of jeans to you? That is a question you can actually answer.

And it undercounts: quitting also frees whatever you were spending on cigarettes, so income rises by more than $100.

PART 5

Demand Elasticities

Two Questions That Need a Number

AT THE AIRPORT AND THE THEATRE

Why is a plane ticket bought the day before so expensive — and a Broadway seat bought that afternoon so cheap?

Same last-minute buyer. Opposite pricing.

AT YOUR GIG

Your band gets the whole cover charge at a bar. You want more money.

Raise the cover and lose some of the crowd, or cut it and pack the room?

Both answers hinge on the same missing number: how much does quantity move when price moves?

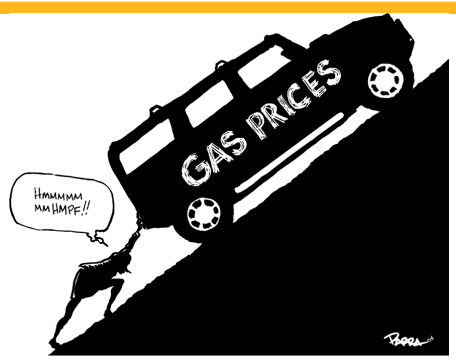
"Quantity falls when price rises" is true and useless here. We need to know by how much.

Elasticity

Elasticity — a measure of how sensitive one variable is to changes in another.

PRICE ELASTICITY OF DEMAND

How much does quantity demanded change when the good's OWN price changes?



The hummer



These are Chapter 4's demand shifters again.

Price Elasticity of Demand

E(D) = percentage change in quantity demanded ÷ percentage change in price

Quarter
Pounder



MINUS SIGN

Demand curves slope down, so E(D) is always negative. By convention we quote the ABSOLUTE VALUE and compare its size to 1.

|E(D)| > 1 elastic · |E(D)| < 1 inelastic · |E(D)| = 1 unit elastic

Percentages on both sides. That is what lets you compare gasoline with concert tickets.

Five Elasticities



PERFECTLY INELASTIC
|E| = 0



INELASTIC
|E| < 1



UNIT ELASTIC
|E| = 1



ELASTIC
|E| > 1

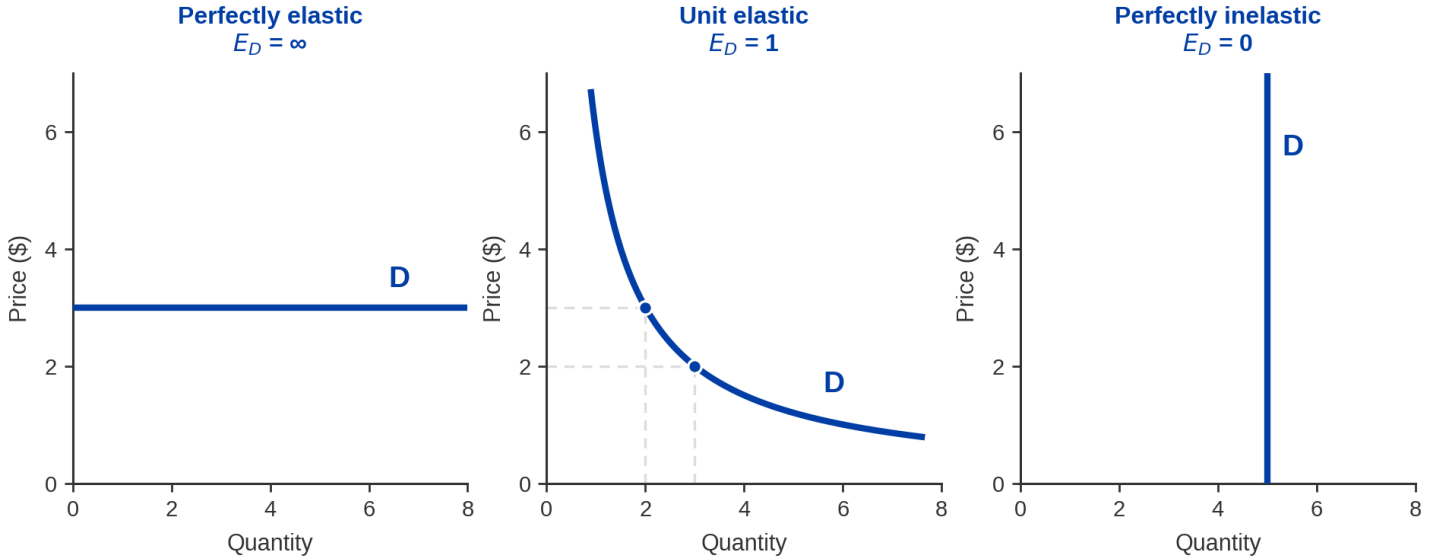


PERFECTLY ELASTIC
|E| = ∞

Insulin sits near the left of this strip — no close substitute, so quantity barely responds. Ketchup sits near the right — dozens of near-identical alternatives on the same shelf.

The two ends are theoretical limits. Everything you will meet in practice lives between them.

Exhibit 5.12 — The Three Extreme Cases



A flat curve means any price rise loses every buyer. A vertical one means the price is irrelevant to quantity.

Exhibit 5.13 — Some Estimated Price Elasticities

Good	|Price elasticity|	Reads as
Olive oil	1.92	elastic
Peanut butter	1.73	elastic
Ketchup	1.36	elastic
Wine	1.00	unit elastic
Laundry detergent	0.81	inelastic
Shampoo	0.79	inelastic
Potato chips	0.45	inelastic
Cigarettes	0.40	inelastic

Elasticity, Measured on Mrs Krank

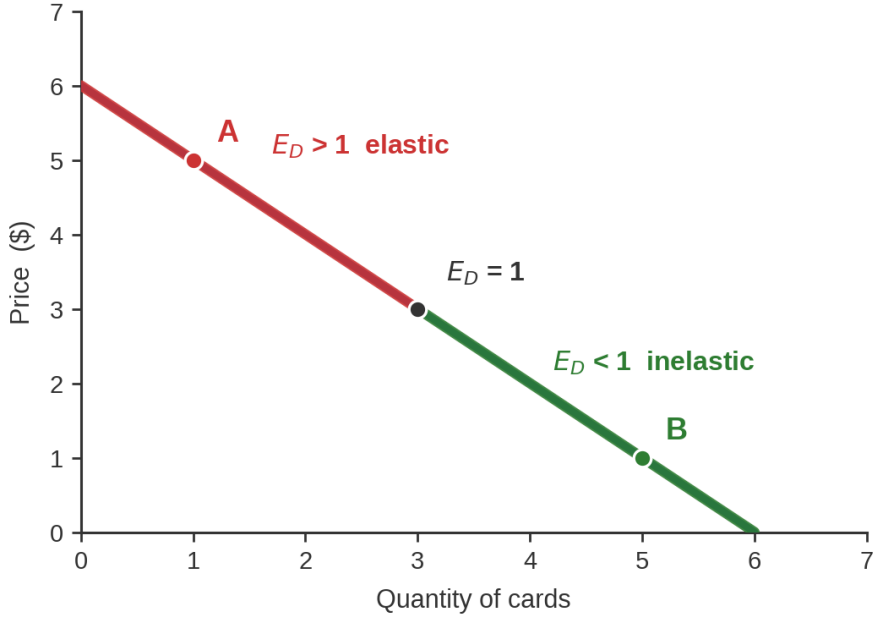
Elasticity — a measure of how sensitive one variable is to changes in another.



Price × 2.5, quantity unchanged: E(D) = 0% ÷ 150% = 0. Perfectly inelastic.

Elasticity is not about how much she likes ham. It is about whether she has anywhere else to go — and tonight she does not.

Exhibit 5.11 — Slope Is Not Elasticity



This is one straight line, so the slope is the same everywhere. But the elasticity is different at every single point.

WHY

Slope compares changes in units. Elasticity compares changes in percentages — and a $1 move is a huge percentage of $1 and a tiny one of $5.

High on the curve, demand is elastic. Low down, it is inelastic. Exactly halfway, it is unit elastic.

The Arc (Midpoint) Formula

arc E(D) = ΔQ ÷ average Q /(ΔP ÷ average P)

THE PROBLEM IT FIXES

$8 → $4 is a 50% cut.
$4 → $8 is a 100% rise.

Same two prices, two different percentages — so two different elasticities for one stretch of curve.

HOW IT FIXES IT

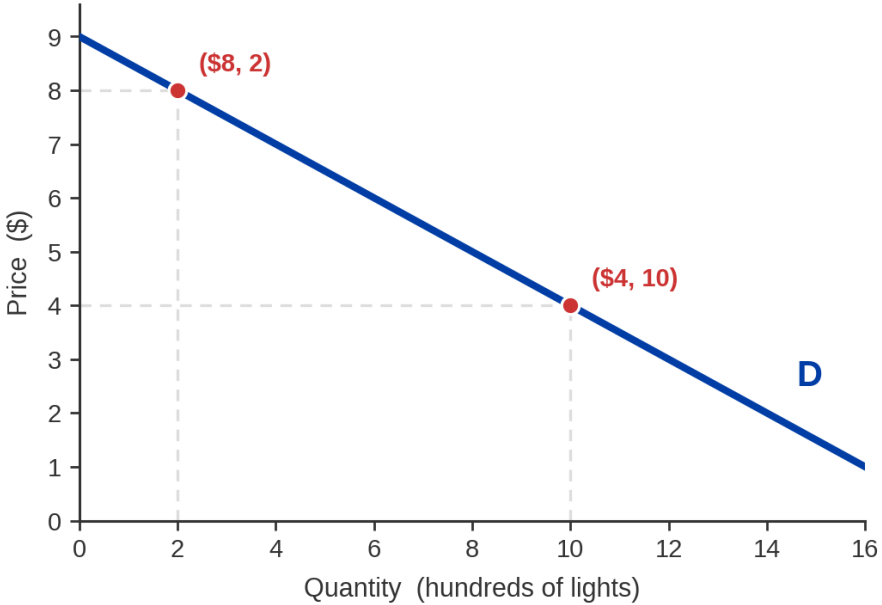
Divide by the AVERAGE of the two prices and the AVERAGE of the two quantities.

Now it does not matter which end you start from. One stretch, one number.

Use the arc formula whenever a question gives you two points. Use the point formula only for small changes.

Direction-free by construction. That is the whole reason it exists.

Practice 1 — Price Falls from $8 to $4



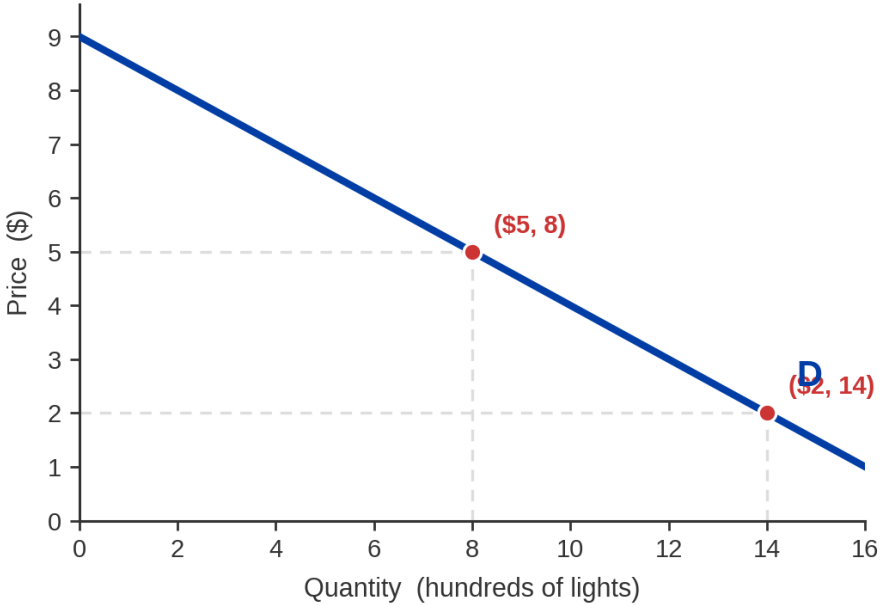
SET IT OUT

Quantity at $8	**200 lights**
Quantity at $4	**1,000 lights**
ΔQ ÷ average Q	**800 ÷ 600 = 1.333**
ΔP ÷ average P	**−4 ÷ 6 = −0.667**

|arc E(D)| = 1.333 ÷ 0.667 = 2.0 — elastic

Averages in both denominators. Start from either end and the answer is the same.

Practice 2 — Price Falls from $5 to $2



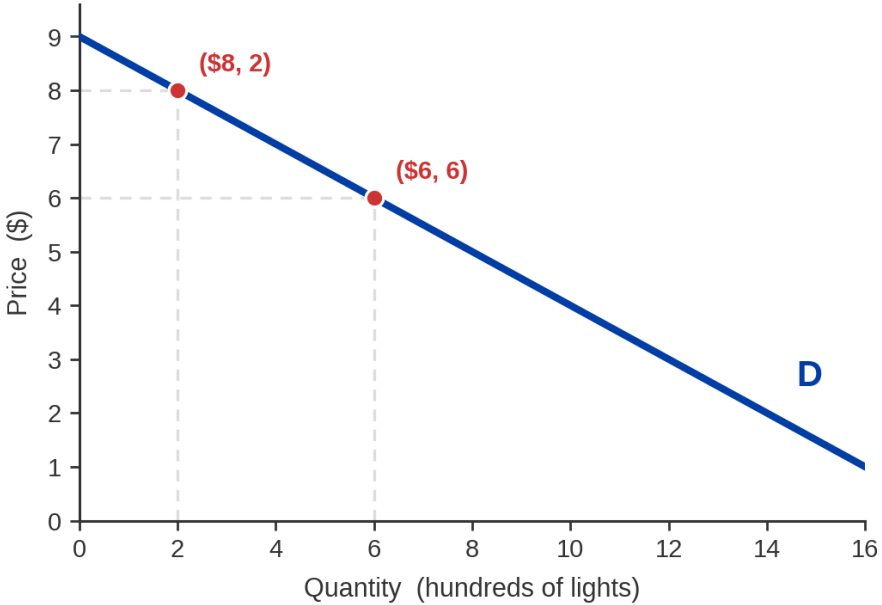
SET IT OUT

Quantity at $5	**800 lights**
Quantity at $2	**1,400 lights**
ΔQ ÷ average Q	**600 ÷ 1,100 = 0.545**
ΔP ÷ average P	**−3 ÷ 3.5 = −0.857**

|arc E(D)| = 0.545 ÷ 0.857 = 0.64 — inelastic

Averages in both denominators. Start from either end and the answer is the same.

Practice 3 — Price Rises from $6 to $8



SET IT OUT

Quantity at $6	**600 lights**
Quantity at $8	**200 lights**
ΔQ ÷ average Q	**−400 ÷ 400 = −1.000**
ΔP ÷ average P	**2 ÷ 7 = 0.286**

|arc E(D)| = 1.000 ÷ 0.286 = 3.5 — strongly elastic

Averages in both denominators. Start from either end and the answer is the same.

Total Revenue: Why Anyone Cares

total revenue = price × quantity

A PRICE RISE PULLS BOTH WAYS

Every unit sold now earns more — revenue **up**.

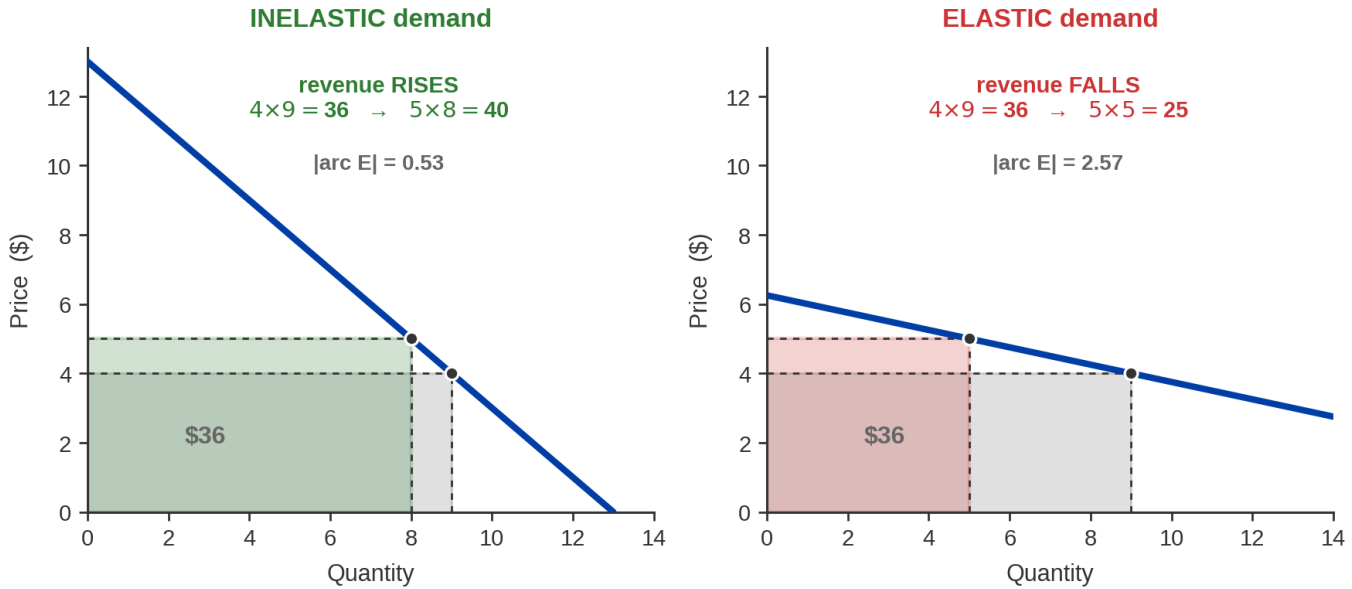
But fewer units are sold — revenue **down**.

ELASTICITY SAYS WHICH WINS

It is a race between a percentage gain in price and a percentage loss in quantity.

Elasticity is precisely the ratio of those two percentages.

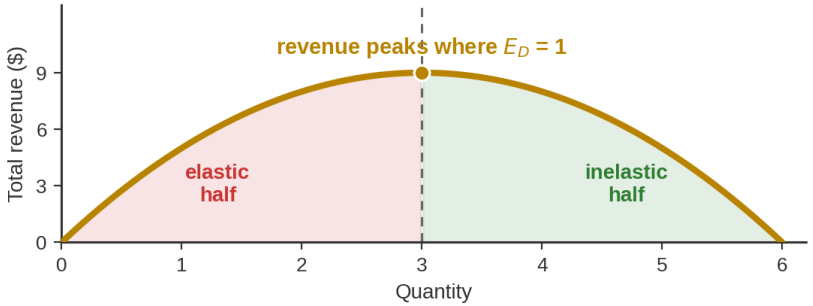
The Same $1 Rise on Two Different Curves



Identical price rise, opposite revenue result. The only thing that differs is how far quantity moved.

The Total-Revenue Rule

If demand is…	Value	Raise the price	Cut the price
Elastic	|E| > 1	revenue FALLS	revenue RISES
Unit elastic	|E| = 1	no change	no change
Inelastic	|E| < 1	revenue RISES	revenue FALLS



IN REVERSE

Raise price, revenue up? Demand is inelastic.

AND THE PEAK

Revenue is highest at |E| = 1 — the midpoint.

So: Should Your Band Raise the Cover Charge?

It depends on who is coming through the door.

FRIENDS AND FAMILY

They came to see YOU. No substitute exists — nobody else is your band.

Demand is inelastic. RAISE the cover; revenue goes up.

STRANGERS ON A FRIDAY NIGHT

They came for a night out. Four other bars are within walking distance.

Demand is elastic. CUT the cover and pack the room.



What Makes Demand Elastic? Three Things

CLOSENESS OF SUBSTITUTES

The big one. If a near-identical alternative sits on the same shelf, buyers leave the moment your price moves.

Ketchup: elastic.
Insulin: not.

SHARE OF THE BUDGET

A 10% rise on a $2 coffee is twenty cents — you barely notice.

A 10% rise on rent is real money, and you will move.

TIME TO ADJUST

Gas doubles tomorrow and you still drive to work.

Give it two years and you have changed car, job or route. Demand gets more elastic with time.

Cross-Price Elasticity of Demand

E(cross) = % change in quantity demanded of good 1 ÷ % change in the price of good 2

Sign	What it means
Negative	Complements
Zero	Unrelated
Positive	Substitutes

Exhibit 5.14 — examples	E(cross)
Meat and fish	1.6
Clothing and entertainment	0.6
Whole milk and low-fat milk	0.5
Meat and potatoes	−0.2
Food and entertainment	−0.7

Here the SIGN is the point, not the size. Positive means buyers switch between them; negative means they are used together.

Chapter 4 told you substitutes and complements exist. This tells you which is which, and how strongly.

Income Elasticity of Demand

E(income) = % change in quantity demanded ÷ % change in income

Value	Type of good
Less than 0	Inferior
Between 0 and 1	Normal — a necessity
Greater than 1	Normal — a luxury

Exhibit 5.15 — selected	E(income)
Foreign vacation	2.10
Healthcare	1.18
Housing	1.00
Gasoline	0.48
Electricity	0.23
Rice	−0.44
Public transit	−0.75

Chapter 4's normal and inferior goods, now with a dividing line at zero — and a second one at 1.

Back to the Airport and the Theatre

THE LAST-MINUTE FLYER

Booking two days out usually means a funeral, a deal, a deadline. There is no substitute for BEING THERE ON TUESDAY.

Demand is inelastic → the airline raises the fare.

A 2024 study of 917 million fares put the 0–6-days-out premium at about 59% above the cheapest booking window.



THE LAST-MINUTE THEATREGOER

Wants a nice evening. A different show, a cinema, a restaurant — all substitute perfectly.

Demand is elastic → the seat gets discounted rather than wasted.

TKTS day-of-show seats were 30–50% off when this was checked in August 2026.



Letting the Data Speak: Should McDonald's Care?

Three questions a burger chain asks — one for each elasticity.

OWN PRICE

Put 30¢ on a burger. Do we make more money or less?

Price elasticity of demand decides it.

CROSS PRICE

Burger King cuts the Whopper price. How many of our customers walk?

Cross-price elasticity.

INCOME

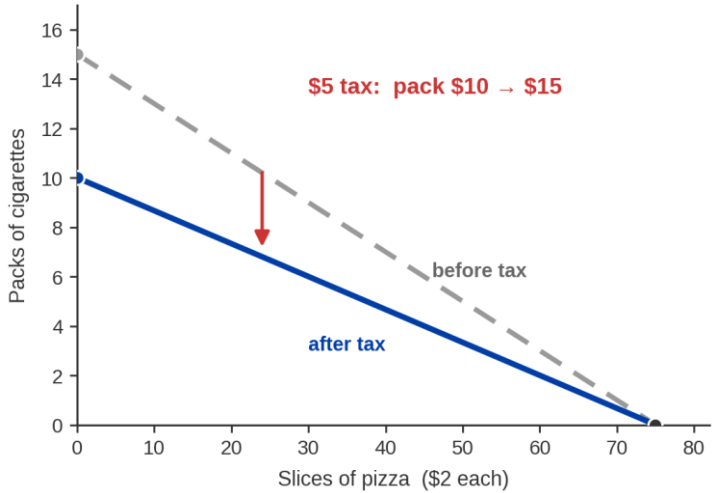
A recession arrives. Do we sell fewer burgers — or more, as people trade down?

Income elasticity.

The category elasticity for food away from home is about 0.81 — but a single BRAND faces far more elastic demand.

Practice:

Greta has $150 to spend on cigarettes and pizza. A pack costs $10; a slice costs $2.



FOUR QUESTIONS

a. Draw Greta's budget constraint.

b. What is her opportunity cost of one pack of cigarettes?

c. Government adds a $5 tax per pack. Draw the new constraint.

d. What is her opportunity cost now?

e. Will she smoke more or less — and which elasticity tells you?

Practice: The Answers

B. BEFORE THE TAX

$10 pack ÷ $2 slice = 5

One pack costs her five slices of pizza. That is the slope of the line.

D. AFTER THE TAX

$15 pack ÷ $2 slice = 7.5

One pack now costs seven and a half slices. Smoking got dearer in pizza.

e. She smokes less. The measure that tells you HOW MUCH less is the PRICE ELASTICITY OF DEMAND.

And because cigarette demand is inelastic, "less" means a little less — which is why the tax raises revenue.

The Whole Chapter, in One Negotiation

A price gets set between Drama and the realtor. Which one it lands nearer is decided by who has somewhere else to go.



BUDGET CONSTRAINT

Drama names a budget and then exceeds it. He ends up in its unaffordable zone.

WHO IS INELASTIC HERE?

Drama is. He wants THIS apartment, and he wants it bad.

SELLER'S MARKET OR BUYER'S?

The realtor doesn’t even flinch as they tries to haggle.

The inelastic side of a bargain pays for the privilege. Mrs Krank paid it for a ham; Drama pays it for a higher quote.

Chapter 5: The Full Picture

	What it says	Key terms	Where you saw it
The buyer's problem	Tastes, prices and a budget. Nothing else is needed to explain a purchase	budget set, budget constraint, revealed preference	Sweaters and jeans on $300
The optimum	Buy in order of marginal benefit per dollar; stop when the ratios are equal	diminishing marginal benefit, consumer equilibrium	6 sweaters, 3 pairs of jeans
Moving the line	A price change pivots the constraint; an income change shifts it	pivot, parallel shift, opportunity cost	Sweaters at $50; income at $600
The demand curve	Re-solve the problem at each price and the curve appears — it is not assumed	law of demand, individual and market demand	Jeans at $25, $50, $75, $100
Consumer surplus	What buyers keep: willingness to pay minus what they actually pay	consumer surplus, willingness to pay	$75 for one buyer, $2.25 bn for the market
Elasticity	How much quantity responds — and therefore what a price change does to revenue	price, cross-price and income elasticity, arc formula, total-revenue rule	Solar lights, the band's cover charge, Greta's tax

Chapter 4 gave you the curve. Chapter 5 gave you the person behind it — and a number for how hard she pushes back.

KEY TAKEAWAYS

Chapter 5: Consumers and Incentives

- The buyer's problem is only three things: what you like, what it costs, what you have
- Buy in order of marginal benefit per dollar, and stop when the ratios are equal
- Re-solve that problem at every price and the demand curve is derived, not assumed
- Consumer surplus is the gap between what you would have paid and what you did pay
- Elasticity turns "quantity responds" into a number — and that number decides revenue